

BIS REFERENCE HANDBOOK: MECHANICAL TESTING & RISK ASSESSMENT

STANDARDS FOR TENSILE, IMPACT, HARDNESS & MACHINERY RISK

Technical manual detailing standardized test procedures for metals, polymers, pressure vessels, and industrial safety machinery.

STANDARDIZED METALLURGICAL TEST METHODS

Test Category	Standard Number	Test Method & Parameters
Tensile Testing of Metals	IS 1608 (Part 1) / ISO 6892-1	Determination of Yield Strength (0.2% Proof Stress), Ultimate Tensile Strength (UTS), % Elongation at ambient temperature
Charpy Impact Test (V-notch)	IS 1757 (Part 1) / ISO 148-1	Measurement of impact energy absorption in Joules (J) at +20°C, 0°C, or sub-zero -40°C temperatures
Rockwell Hardness Test	IS 1586 (Part 1) / ISO 6508-1	HRA, HRB, HRC scales using diamond spheroconical indenter or tungsten carbide ball
Vickers & Brinell Hardness	IS 1501 & IS 1500	Micro and macro hardness testing under loads from 1 kgf to 3000 kgf
Machinery Risk Assessment	IS/ISO 12100 : 2010	Risk assessment and risk reduction principles for industrial machinery